PSYCH/COMP SCI/BMI 841: Computational Cognitive Science

Fall 2025

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Class Presentation Guidelines

Each student is responsible for two class presentations, which present and guide discussion on a recent research paper in human and machine learning. You should read the paper carefully and primarily focus on how the mathematical models connect with cognitive science, and the evidence provided by the authors for the appropriateness of the connection.

Before your scheduled presentation, **I recommend meeting with me during office hours (or a scheduled time) to discuss your presentation**. Most research papers have more information than you could discuss in class, and so we will discuss the aspects your presentation should focus on. Each presentation should be **no longer than 20 minutes in total**. Your presentation of the article itself should end with questions to guide class discussion about the paper. Have at least 3 questions with a goal of generating 10-15 minutes of class discussion.

Your grade will be determined by the following factors:

Understanding of the paper (1.5 points)
Covering the key aspects of the paper (1.5 points)
Presentation clarity (1 points)
Appropriate discussion questions (0.5 point)
Appropriate use of time (0.5 points)

Here are some useful tips for making effective presentations (adapted from Tom Griffiths).

For class presentations, focus on the following questions:

1. What is the cognitive science question that this paper explores? What aspect of human behavior are we trying to understand?
2. What is the context of the paper? What previous approaches have been used to investigate this question, if any?
3. What is the underlying computational problem being solved? How does the paper formulate this problem?
4. What is the solution to this problem presented in the paper? How can you explain the basic mathematical ideas clearly? (Focus on intuitions more than rigor.)
5. How is this solution used to understand the investigated aspect of human behavior? How does the model compare to human performance?
6. What conclusions can we draw from these results? What is your interpretation?

You can use these questions to organize your presentation. They are also useful to keep in mind when reading papers for the class.

One grading criterion is presentation clarity. Included in this is the effectiveness of the visual aspects of the presentation. Effectiveness is key. It is not about having fancy PowerPoint transitions or using fancy presentation software. In fact, both of these are actively discouraged. (E.g., I dislike Prezi because I find it extremely distracting and it makes me nauseous.). You should be as succinct as possible, including the minimum amount of text that still makes your presentation comprehensible.

Other tips:

1. Not every point has to be its own slide filled with text.

2. Presenting equations is fine, but please make sure you explain what the equation means.
3. Using graphs from papers to discuss results is encouraged. But, make sure your axes are labeled, the lines are legible, and that it has a title. Also, make sure you clearly explain what is being plotted and why the graph is important. Understanding what information the graph conveys is more important than noting the p-value of some difference without any context.
4. Try not to use any text smaller than 24pt font.
5. Take a break and a deep breath every few slides. Talk slower than you normally do during a conversation.
6. Although the maximum length of time for a presentation is 25, you do not need to use all of the time. If you can present the important details in a shorter amount of time, that is great! Just don’t take too long. There should be time for discussion and for the other presentations scheduled that day.
7. Practice your presentation out loud at least once. It is especially effective if you can convince a friend to listen to your presentation.